

CHEMICAL REACTOR DIGITAL TWIN

Vendor-neutral, calibratable R&D reference | Open source MIT | vicena.ai

Batch, CSTR, PFR

Mass + energy balances

Kinetics calibration

Identifiability diagnostics

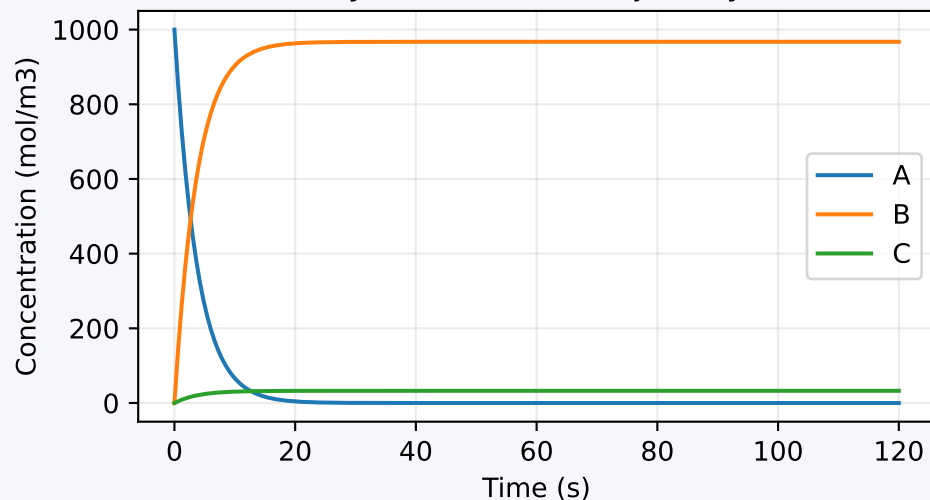
Validation and uncertainty

Held-out metrics

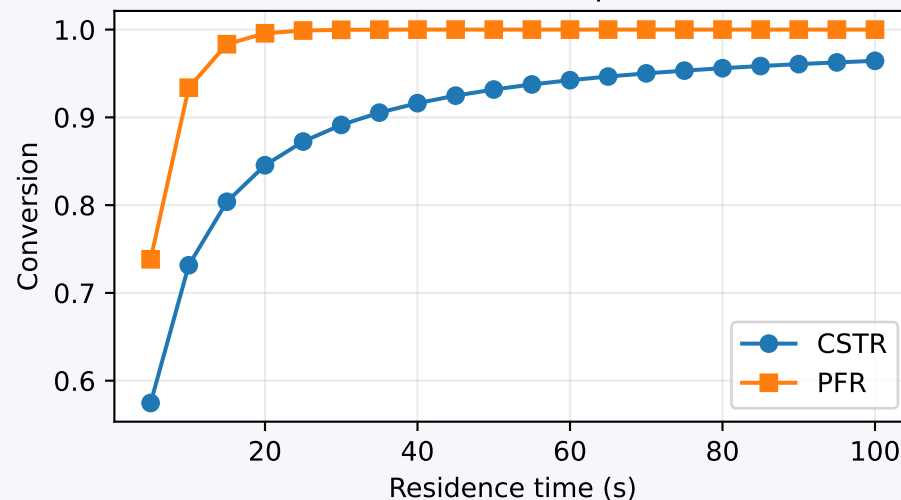
Constrained optimization

Research hypotheses only

Synthetic batch trajectory



Ideal reactor comparison



Synthetic baseline: batch conversion 1.000 | B selectivity 0.967 | peak T 334.47 K | solver BDF

1 Validate data | 2 Simulate | 3 Calibrate | 4 Hold-out validate | 5 Analyze and optimize safely